Multiple Regression Model to Analyze the Effect of Socioeconomic Factors on Cancer Mortality Rate in U.S. Counties

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Fall 2025

Contents

1 Introduction

Cancer is a disease that is becoming increasingly prevalent in our aging society. The progression and incidence of this disease of aging are associated with a wide array of factors including genetic risk factors, environmental carcinogens, and the interaction between the two.[1,2] Environmental carcinogens include specific toxins, stress, sleep disruption, and many more.[3] Exposure to these factors can not only increase the likelihood of cancer incidence, but accelerate carcinogenesis in patients who have cancer already.[4] The exposure of individuals to specific carcinogens can be influenced by socioeconomic factors. As such, socioeconomic status has been shown to be a predictor of a variety of health and illness outcomes.[5]

The goal of this study is to investigate the effects of various socioeconomic factors on cancer mortality in various U.S. counties. In this study, a predictive model of the rate of cancer mortality per 100,000 residents of various U.S. counties was constructed using multiple linear regression to investigate this relationship.

2 Data

The data set used in this study was aggregated from several sources: the 2013 United States Census, the National Library of Medicine - Clinical Trials, and the National Cancer Institute. This data set included the information of $n = 3048$ U.S. Counties over 7 years (2010-2016).[6] The response variable of interest was the rate of Cancer Mortality per 100,000 residents and the socioeconomic predictor variables investigated in this data set and their features are shown in Table 1.

Table 1: Descriptive Statistics of Socioeconomic Factors and Cancer Mortality

Variable	$\bar{\mathbf{x}}$	$\tilde{\mathbf{x}}$	**Min**	**Max**
Response Variable				
Cancer Mortality Rate per Capita	178.7	178.1	59.7	362.8
Predictor Variables				
Cancer Incidence Rate	448.3	453.5	201.3	1206.9
Median Income	47063.3	45207.0	22640.0	125635.0
Population	102637.4	26643.0	827.0	10170292.0
Percent of Populace in Poverty	16.9	15.9	3.2	47.4
Number of Cancer-Related Clinical Trials per Capita	155.4	0.0	0.0	9762.3
Median Age of Residents	45.3	41.0	22.3	624.0
Median Age of Male Residents	39.6	39.6	22.4	64.7
Median Age of Female Residents	42.1	42.4	22.3	65.7
Mean Household Size	2.5	2.5	0.0	4.0
Percent of Residents who are Married	51.8	52.4	23.1	72.5
Percent of Married Households	51.2	51.7	23.0	78.1
Birth Rate	5.6	5.4	0.0	21.3
Percent of Residents Aged 18-24 Highest Education: less than High School	18.2	17.1	0.0	64.1
Percent of Residents Aged 18-24 Highest Education: High School Diploma	35.0	34.7	0.0	72.5
Percent of Residents Aged 18-24 Highest Education: Some College*	41.0	40.4	7.1	79.0
Percent of Residents Aged 18-24 Highest Education: Bachelor's Degree	6.2	5.4	0.0	51.8
Percent of Residents Over Age 25 Highest Education: High School Diploma	34.8	35.3	7.5	54.8
Percent of Residents Over Age 25 Highest Education: Bachelor's Degree	13.3	12.3	2.5	42.2
Percent of Employed Residents Over Age 16*	54.2	54.5	17.6	80.1
Percent of Unemployed Residents Over Age 16	7.9	7.6	0.4	29.4
Percent of Residents with Private Health Coverage	64.4	65.1	22.3	92.3
Percent of Residents with Private Health Coverage Alone (no public assistance)*	48.5	48.7	15.7	78.9
Percent of Residents with Employee-Provided Private Health Coverage	41.2	41.1	13.5	70.7
Percent of Residents with Government-Provided Health Coverage	36.3	36.3	11.2	65.1
Percent of Residents with Government-Provided Health Coverage Alone	19.2	18.8	2.6	46.6
Percent of Residents who Identify as White	83.6	90.1	10.2	100.0
Percent of Residents who Identify as Black	9.1	2.2	0.0	85.9
Percent of Residents who Identify as Asian	1.3	0.5	0.0	42.6
Percent of Residents who Identify as a Race Other than White, Black, or Asian	2.0	0.8	0.0	41.9

* These predictor variables were removed from the analysis due to missing values

This data set was not complete, so the Percent of County Residents Aged 18-24 Highest Education: Some College, Percent of Employed County Residents Over Age 16, and Percent of County Residents with

Private Health Coverage Alone variables were removed as they were missing values. Also, there are some unexpected high values in the Median Age of Residents, possibly due to data entry error.

The distribution of the response variable, Cancer Mortality Rate per 100,000 Residents, is shown in Figure 1 and appears to be normally distributed. Most values lie between 100-250 cancer mortalities per 100,000 people, with a few values above 350 cancer mortalities per 100,000 people. The average Cancer Mortality Rate across the U.S. counties included in this study is 178.1 cancer mortalities per 100,000 people.

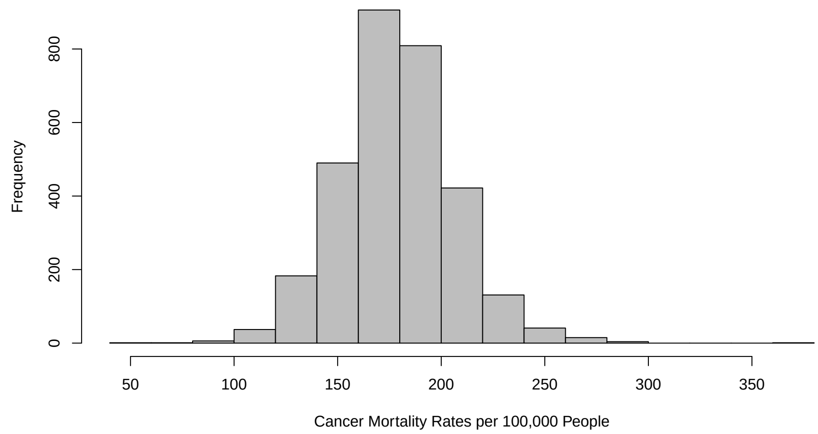

Figure 1: Histogram of Cancer Mortality Rates in U.S. Counties

2.1 Exploratory Data Analysis

The correlation heatmap generated in R shows the relationship between the response and predictor variables in Figure 2:

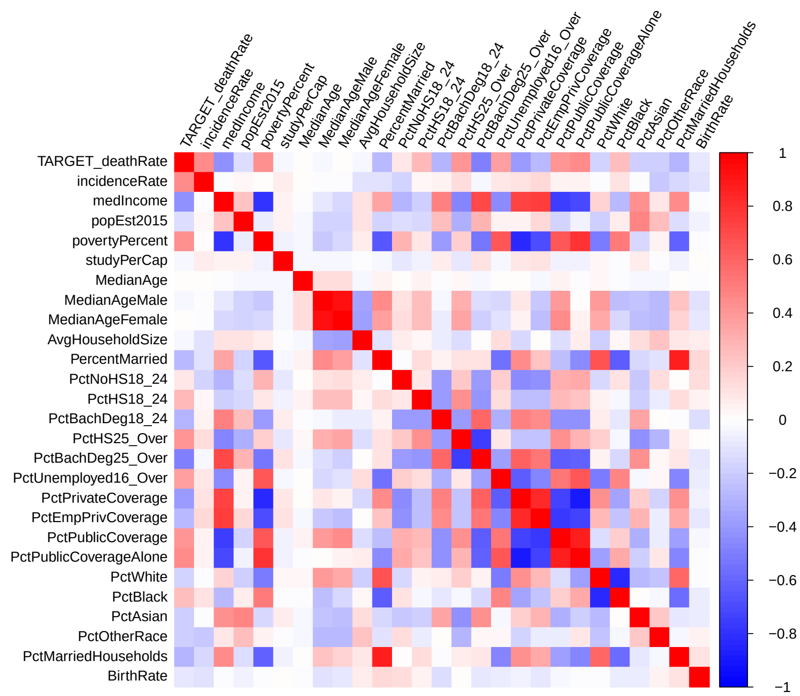

Figure 2: Correlation Heatmap of Socioeconomic Factors and Cancer Mortality

Based on the correlation heatmap, the predictor variables that have the strongest linear relationship with Cancer Mortality Rate are Percent of Residents Over Age 25 Highest Education: Bachelor's Degree ($r = -0.49$), Percent of Residents with Government-Provided Health Coverage Alone ($r = 0.45$), Cancer Incidence Rate ($r = 0.45$), Median Income ($r = -0.43$), and Percent of Populace in Poverty ($r = 0.43$). Figure 3 shows a scatterplot matrix showing the relationships between each predictor variable and between the response variable.

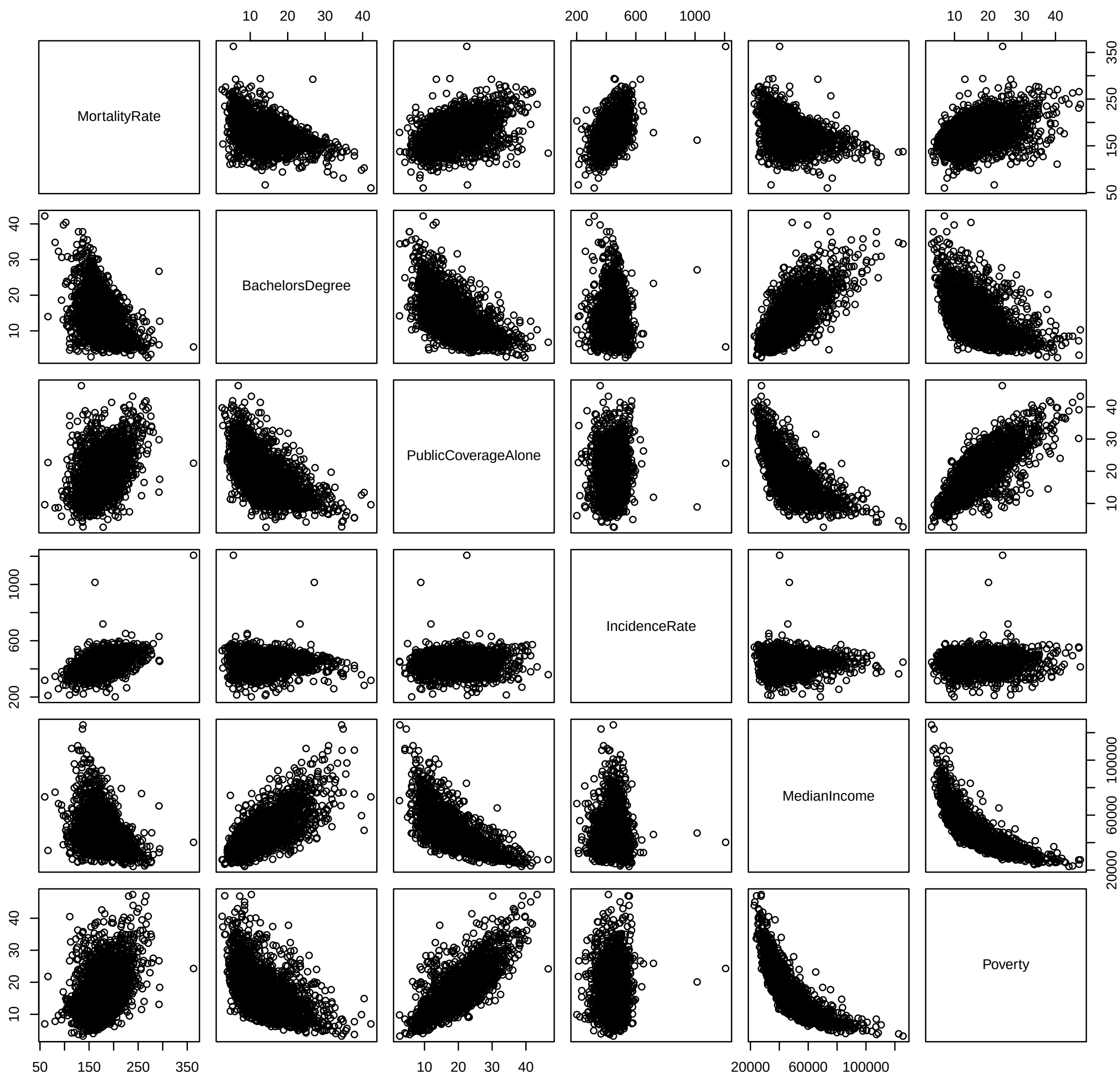

Figure 3: Scatterplot Matrix of Cancer Mortality Rate, Percent of Residents Over Age 25 Highest Education: Bachelor's Degree, Percent of Residents with Government-Provided Health Coverage Alone, Cancer Incidence Rate, Median Income, and Percent of Populace in Poverty

Other than Cancer Incidence Rate, the remaining predictor variables presented in Figure 3 appear to have at least somewhat linear relationships with each other. This could indicate multicollinearity and a formal inspection was performed after creating the regression model.

3 Methods

In this study, the influence of many predictor variables is considered in how they simultaneously impact the Cancer Mortality Rate per Capita in U.S. Counties. Since more than one predictor variable is investigated, the Multiple Linear Regression method is used to fit a model of the form:[7]

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \varepsilon_i \tag{1}$$

where Y_i is the Cancer Mortality Rate per Capita in U.S. Counties, β_0 is the intercept value, x_{ij} are the predictor variables listed in Table 1, β_{ij} is the change in the mean of Y for a unit change in the respective x_{ij} with all other variables held constant, and ε_i is the random error. The assumptions of the Multiple Linear Regression Model are:

- The predictor variables (x's) are fixed or conditioned upon
- The Y_i have a common variance σ^2, which is the same for all values of x
- The errors are normally distributed
- The errors are independent
- The parameters (β_k) have a linear relationship

The Least Squares (LS) fit was used to determine the best fit of the multiple linear regression model to this data set. The LS estimates minimize the sum of the vertical distance of all data points to the fitted line (Q)

$$Q = \sum_{i=1}^{n} [y_i - (\beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik})]^2 \tag{2}$$

to find the estimated LS regression surface:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_k x_k \tag{3}$$

The Goodness of Fit of the LS model was measured using the coefficient of multiple determination (r^2) which represents the proportion of variation in the y_i's that is accounted for by regression on $x_1, x_2, \ldots, x_k$.

$$r^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST} \tag{4}$$

Where SSE is the error sum of squares, SST is the total sum of squares which captures the error when fitting the simplest model ($Y_i = \beta_0 + \varepsilon_i$), and SSR is the regression sum of squares which represents the amount of variation in the response variable that is explained by the predictor variables in the model.

$$SSE = Q_{min} \tag{5}$$

$$SST = \sum (y_i - \hat{y})^2 \tag{6}$$

$$SSR = SST - SSE \tag{7}$$

For testing the statistical significance of the effects of individual predictor variables, the null hypothesis (H_0) indicates none of the predictor variables are related to y and the alternate hypothesis (H_1) indicates at least one of the predictor variables is useful in predicting y or that the model has some ability to predict the variability observed in y.

In this study the significance level ($\alpha = 0.05$) so H_0 is rejected if:

$$F > f_{k,n-k-1,\alpha} \tag{8}$$

or

$$\text{p-value} = P(F_{k,n-k-1} > F) < \alpha \tag{9}$$

These values can also be found using an analysis of variance (ANOVA) table when comparing the full and reduced models.

3.1 Model Diagnostics

To assess the validity of the model and adherence to the assumptions of the Multiple Linear Regression Model method, the following checks are used:

- Check linearity
  - Plot residuals vs. individual predictor variables
  - Remedy: transformation of Y or adjusting predictor variables
- Check constant variance
  - Plot residuals vs. fitted values from LS estimation
  - Remedy: transformation of Y
- Check normality
  - Plot Q-Q plot (normal probability plot)
  - Remedy: transformation of Y or use more robust methods
- Check outliers and influential observations
  - Plot residuals or studentized residuals and find influential observations
  - Remedy: remove outliers/influential observations or use more robust methods
- Check for multicollinearity
  - Find VIF of each predictor variable
  - Remedy: drop variables that are highly correlated

3.1.1 Outliers and Influential Observations

The studentized residuals (e_i^*) were calculated by:

$$e_i^* = \frac{e_i}{\sqrt{MSE\left(1 - \frac{1}{n} - \frac{(x_i - \bar{x})^2}{\sum_i (x_i - \bar{x}^2)}\right)}} \tag{10}$$

The leverage values (h_{ii}) to determine influential observations in the data set were calculated by:

$$\sum_{i=1}^{n} h_{ii} = k + 1 \tag{11}$$

where k is the number of predictor variables.

If a value in the data set had both $|e_i^*| > 3$ and $h_{ii} > 2(k+1)/n$, they were considered an outlier.

3.1.2 Multicollinearity

The Variance Inflation Factor (VIF) was calculated for each predictor variable to determine the degree of multicollinearity to see if a predictor variable significantly increased the variance of $\hat{\beta}_j$.

$$VIF_j = (1 - r_j^2)^{-1} \tag{12}$$

A variable with $VIF > 10$ is considered problematic and variable removal should be considered.

3.2 Software and Tools

All statistical analyses, model fitting, visualizations, and calculations were performed using **R** version 4.3.1. in RStudio.

4 Results and Discussion

4.1 Initial Model

The initial multiple linear regression model for the data set showed that a significant amount of the variance in the Rate of Cancer Mortalities in U.S. Counties could be explained by the socioeconomic determinants in the data set as shown by the coefficient $\hat{\beta}$'s in Table 2.

Table 2: Summary of coefficients $\hat{\beta}$'s of the predictor variables and their significance.

	Estimate ($\hat{\beta}$)	Std. Error	t value	Pr(>|t|)
(Intercept)	124.369	13.235	9.397	**1.1E-20***
Cancer Incidence Rate	0.192	0.007	26.657	**8.9E-141***
Median Income	9.2E-05	8.1E-05	1.135	0.256
Population	-1.9E-06	1.3E-06	-1.462	0.144
Percent of Populace in Poverty	0.562	0.153	3.673	**2.4E-04***
Number of Cancer-Related Clinical Trials per Capita	-2.1E-04	0.001	-0.310	0.757
Median Age	-0.003	0.008	-0.419	0.675
Median Age of Male Residents	-0.501	0.209	-2.395	**0.017***
Median Age of Female Residents	0.057	0.215	0.265	0.791
Mean Household Size	0.475	0.963	0.493	0.622
Percent of Residents who are Married	0.812	0.147	5.526	**3.5E-08***
Percent of Married Households	-0.879	0.140	-6.278	**3.9E-10***
Birth Rate	-0.982	0.194	-5.069	**4.2E-07***
Percent of County Residents Aged 18-24 Highest Education: less than High School	-0.133	0.056	-2.371	**0.018***
Percent of County Residents Aged 18-24 Highest Education: High School Diploma	0.254	0.049	5.155	**2.7E-07***
Percent of County Residents Aged 18-24 Highest Education: Bachelor's Degree	-0.080	0.109	-0.739	0.460
Percent of County Residents Over Age 25 Highest Education: High School Diploma	0.413	0.097	4.263	**2.1E-05***
Percent of County Residents Over Age 25 Highest Education: Bachelor's Degree	-1.191	0.154	-7.733	**1.4E-14***
Percent of Unemployed County Residents Over Age 16	0.451	0.161	2.812	**0.005***
Percent of County Residents with Private Health Coverage	-0.543	0.131	-4.150	**3.4E-05***
Percent of County Residents with Employee-Provided Private Health Coverage	0.337	0.103	3.265	**0.001***
Percent of County Residents with Government-Provided Health Coverage	0.032	0.218	0.145	0.885
Percent of County Residents with Government-Provided Health Coverage Alone	0.068	0.276	0.248	0.804
Percent of County Residents who Identify as White	-0.096	0.057	-1.664	**0.096 .**
Percent of County Residents who Identify as Black	-0.025	0.056	-0.455	0.649
Percent of County Residents who Identify as Asian	-0.005	0.189	-0.029	0.977
Percent of County Residents who Identify as a Race Other than White, Black, or Asian	-0.911	0.125	-7.303	**3.6E-13***

Where level of significance is indicated by: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

4.1.1 Goodness of Fit of the Initial Model

Table 3: Initial Multiple Linear Regression Model Summary and ANOVA Test Comparison with the Reduced Model

Source	r^2	Adjusted r^2	Sum of Squares	d.f.	MSE	F-statistic	p-value
Regression	0.5136	0.5094	1204859	26	46340.7	122.65	<2.2E-16
Error			1141007	3020	377.8		
Total			2345866	3046			

As shown in Table 3, the initial model is able to contribute significantly to the prediction of Cancer Mortality Rate (p-value = $<$ 2.2E-16) in comparison to the reduced model that contains no predictor variables.

4.1.2 Model Diagnostics of the Initial Model

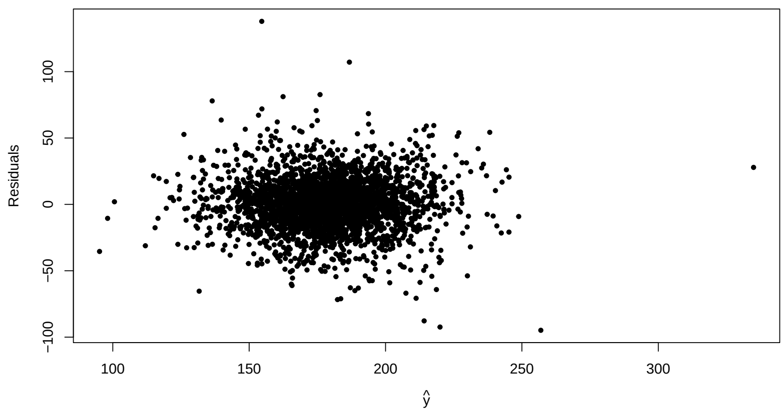

Figure 4: Scatter Plot of Residuals vs. Fitted Values ($\hat{y}$)

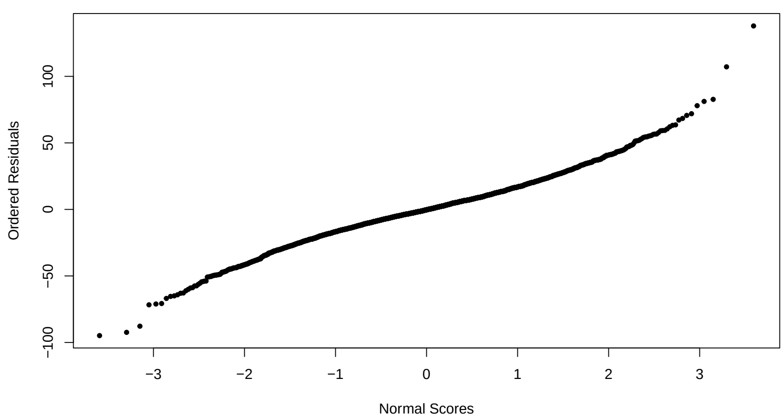

Figure 5: Normal Plot of Residuals

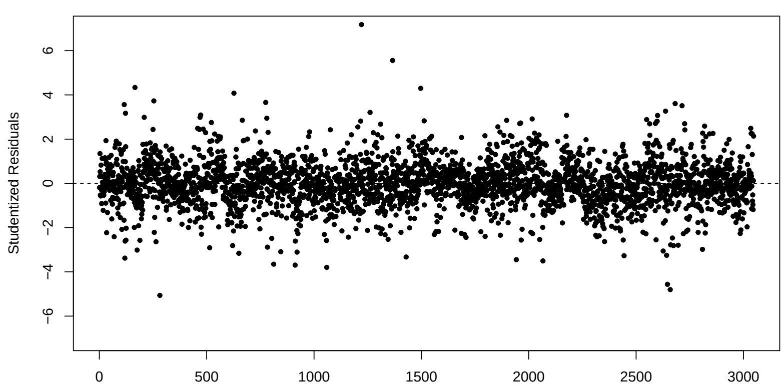

Figure 6: Scatter Plot of Studentized Residuals

Based on the model diagnostics, the variance of the residuals appears to be constant (Figure 4) and approximately normal (Figure 5), the relationship of the response and predictor variables appear to be linear in the parameters (Figure 4), and there are some outliers that need to be further investigated (Figure 6).

4.2 Fitting the Model

In order to improve the model, the Median Age of County Residents predictor variable was removed as the range of the data values were unexpected and unrealistic (Maximum value of 624), possibly due to data entry error. Additionally, the contribution of the Median Age of County Residents predictor variable to the initial model was not significant (p-value = 0.68) as shown in Table 2. The effect of removing Median Age of County Residents on the coefficents of the predictor variables was not significant and is shown in Table 8 in the Appendix.

After the the Median Age of County Residents predictor variable was removed, the remaining predictor variables were checked for multicollinearity. The largest VIF value was found to be Percent of County Residents with Government-Provided Health Coverage (23.58). Percent of County Residents with Government-Provided Health Coverage did not contribute significantly to the model (p-value = 0.88) and was removed. The model was re-evaluated after removing Percent of County Residents with Government-Provided Health Coverage as a predictor variable as shown in Table 9 in the Appendix. The most significant change was in the Percent of County Residents with Government-Provided Health Coverage Alone as shown in Table 4.

Table 4: Summary of coefficients ($\hat{\beta}$'s) of the predictor variables and their significance after removing Percent of County Residents with Government-Provided Health Coverage predictor variable.

	Estimate ($\hat{\beta}$)	Std. Error	t value	Pr(>|t|)
Before Removing				
Percent of County Residents with Government-Provided Health Coverage	0.033	0.218	0.151	0.8803
Percent of County Residents with Government-Provided Health Coverage Alone	0.069	0.276	0.250	0.8025
After Removing				
Percent of County Residents with Government-Provided Health Coverage Alone	0.104	0.148	0.701	0.4836

Where level of significance is indicated by: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The decrease in p-value and $SE(\hat{\beta}_j)$ indicates that multicollinearity was correctly identified and remedied. The estimates ($\hat{\beta}$) of the remaining predictor variables shown in Table 9 did not change significantly.

After the Percent of County Residents with Government-Provided Health Coverage predictor variable was removed, the VIF of the remaining predictor variables was calculated again to see if there were more variables that could have multicollinearity. The only VIF greater than 10 was for Percentage of County Residents with Private Health Coverage ($VIF = 12.36$). However, the contribution of this predictor variable was significant (p-value = 4.5E-6, Table 9 so the Percentage of County Residents with Private Health Coverage predictor variable was not removed.

Once multicollinearity was remedied and predictor variables with unrealistic values was removed, values in the data set that were both outliers based on the studentized residuals ($|e_i^*| > 3$) and leverage values ($h_{ii} > 2(k+1)/n$), were considered an outlier and removed. 15 outliers were detected and removed. The predictor variables of the linear model with outliers removed (Table 5) is able to contribute significantly to the prediction of Cancer Mortality Rate in comparison to the reduced model that contains no predictor variables as shown in Table 6.

Table 5: Summary of coefficients ($\hat{\beta}$'s) of the predictor variables expected to change and their significance after removing Median Age of County Residents and Percentage of County Residents with Government-Provided Health Coverage.

	Estimate ($\hat{\beta}$)	Std. Error	t value	Pr(>|t|)
(Intercept)	125.367	13.029	9.623	1.3E-21
Cancer Incidence Rate	0.196	0.007	27.164	**1.7E-145** ***
Median Income	0.000	0.000	1.756	**0.079 .**
Population	-0.000	0.000	-1.224	0.221
Percent of Populace in Poverty	0.680	0.149	4.562	**5.3E-06** ***
Number of Cancer-Related Clinical Trials per Capita	-0.000	0.001	-0.570	0.568
Median Age of Male County Residents	-0.658	0.206	-3.202	**0.001** **
Median Age of Female County Residents	0.166	0.199	0.834	0.404
Mean Household Size	-0.149	0.956	-0.156	0.876
Percent of Residents who are Married	0.906	0.146	6.208	**6.1E-10** ***
Percent of Married Households	-0.908	0.138	-6.572	**5.8E-11** ***
Birth Rate	-1.006	0.191	-5.260	**1.5E-07** ***
Percent of County Residents Aged 18-24 Highest Education: less than High School	-0.115	0.055	-2.092	**0.037** *
Percent of County Residents Aged 18-24 Highest Education: High School Diploma	0.239	0.048	4.994	**6.3E-07** ***
Percent of County Residents Aged 18-24 Highest Education: Bachelor's Degree	-0.025	0.105	-0.236	0.813
Percent of County Residents Over Age 25 Highest Education: High School Diploma	0.386	0.095	4.082	**4.6E-05** ***
Percent of County Residents Over Age 25 Highest Education: Bachelor's Degree	-1.188	0.149	-7.999	**1.8E-15** ***
Percent of Unemployed County Residents Over Age 16	0.578	0.156	3.714	**2.1E-04** ***
Percent of County Residents with Private Health Coverage	-0.614	0.115	-5.322	**1.1E-07** ***
Percent of County Residents with Employee-Provided Private Health Coverage	0.384	0.092	4.189	**2.9E-05** ***
Percent of County Residents with Government-Provided Health Coverage Alone	0.057	0.146	0.391	0.696
Percent of County Residents who Identify as White	-0.123	0.057	-2.163	**0.031** *
Percent of County Residents who Identify as Black	-0.065	0.055	-1.180	0.238
Percent of County Residents who Identify as Asian	-0.259	0.192	-1.346	0.178
Percent of County Residents who Identify as a Race Other than White, Black, or Asian	-1.012	0.122	-8.288	**1.7E-16** ***

Where level of significance is indicated by: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

4.2.1 Goodness of Fit of the Fitted Model

Table 6: Multiple Linear Regression Model Summary with Outliers, Median Age of County Residents and Percentage of County Residents with Government-Provided Health Coverage Predictor Variables Removed and ANOVA Test Comparison with the Reduced Model

Source	r^2	Adjusted r^2	Sum of Squares	d.f.	MSE	F-statistic	p-value
Regression	0.5350	0.5313	1228874	24	51203.08	144.16	$< 2.2E-16$
Error			1067996	3007	355.17		
Total			2296870	3031			

In comparison to the initial model, the fitted model displays a greater Goodness of Fit and greater simplicity due to fewer predictor variables.

4.2.2 Model Diagnostics of the Fitted Model

After the model was fitted, the model diagnostics show that assumptions are similarly followed. The variance of the residuals appears to be constant (Figure 7) and normal (Figure 8), the relationship of the response and predictor variables appear to be linear in the parameters (Figure 7), and there do not appear to be extreme outliers that affect the model (Figure 9)

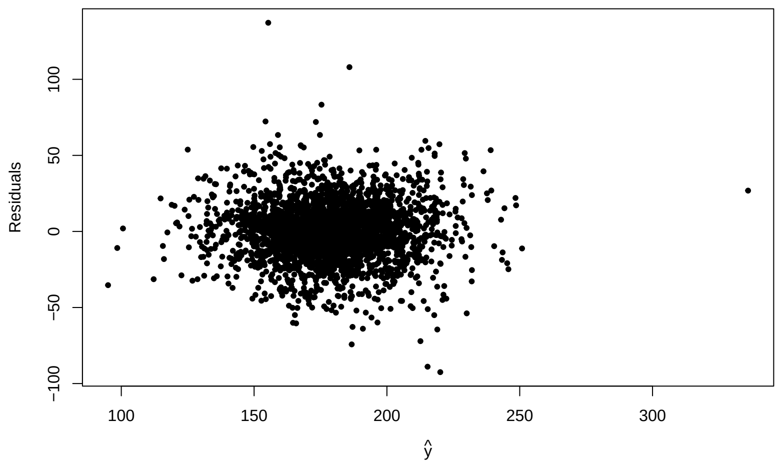

Figure 7: Scatter Plot of Residuals vs. Fitted Values ($\hat{y}$)

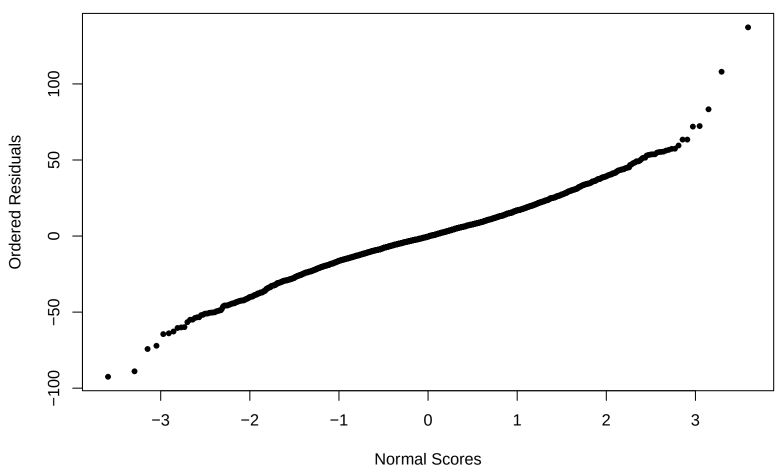

Figure 8: Normal Plot of Residuals

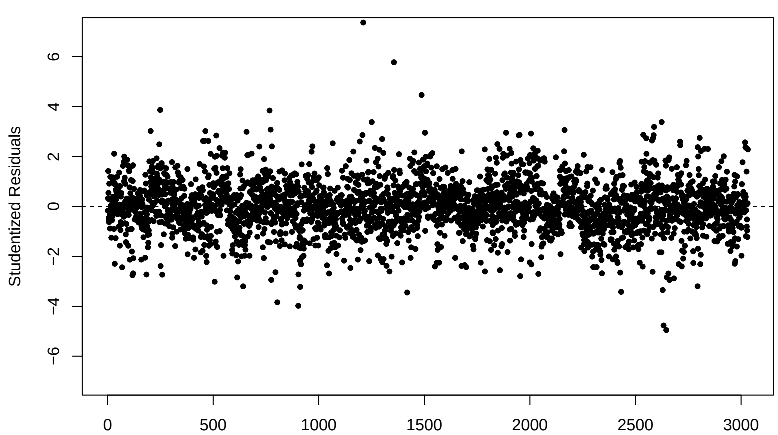

Figure 9: Scatter Plot of Studentized Residuals

4.3 Statistical Inference on Coefficient β's

To see if the overall model is significant at the $\alpha = 0.05$ significance level, the r^2 value, F-statistic, and p-value from Table 6 are evaluated with the following hypotheses:

- H_0: all $\beta_j = 0$; indicates none of the predictor variables are related to y
- H_1: at least one $\beta_j \neq 0$; indicates at least one of the predictor variables is useful in predicting y or that the model has some ability to predict the variability observed in y

The p-value of the overall model was very small ($< 2.2E - 16$) so we reject the null hypothesis and determine that the model is significant in predicting some of the variability observed in the Cancer Mortality Rate per 100,000 Residents in U.S. Counties. Since $r^2 = 0.5350$, about 54% of the total variance in the response variable, Cancer Mortality Rate per 100,000 Residents in U.S. Counties, is explained by the predictor variables.

To see if the effects of each individual predictor variables are significant at the $\alpha = 0.05$ significance level, the p-values of each predictor variable in the fitted linear model was evaluated with the following hypotheses:

- H_0: $\beta_j = 0$; indicates the predictor variables is not related to y
- H_1: $\beta_j \neq 0$; indicates the predictor variables is useful in predicting y, holding all other predictor variables constant

The significance of the individual predictor variables was further evaluated with the construction of 95% confidence intervals. To determine the true coefficient for each predictor variable, 95% Confidence Intervals were built for each predictor variable from Table 5 and shown in Table 7. We are 95% confident that the true coefficient for each predictor variable lies between the lower and upper bounds shown below. If the interval does not include 0, we can say with 95% confidence that the predictor variable has some effect on the Cancer Mortality Rate, all other predictor variables held constant.

The following predictor variables have a p-value $< \alpha = 0.05$ and have 95% confidence intervals that do not include 0 and are therefore concluded to contribute significantly to the linear model:

- Cancer Incidence Rate
- Percent of Populace in Poverty
- Median Age of Male County Residents
- Percent of Married Individuals
- Percent of Married Households
- Birth Rate
- Percent of County Residents Aged 18-24 Highest Education: less than High School
- Percent of County Residents Aged 18-24 Highest Education: High School Diploma
- Percent of County Residents Over Age 25 Highest Education: High School Diploma
- Percent of County Residents Over Age 25 Highest Education: Bachelor's Degree
- Percent of Unemployed County Residents Over Age 16
- Percent of County Residents with Private Health Coverage
- Percent of County Residents with Employee-Provided Private Health Coverage
- Percent of County Residents who Identify as White
- Percent of County Residents who Identify as a Race Other than White, Black, or Asian

Table 7: 95% Confidence Intervals for Coefficients (β_j)

Predictor	**2.5%**	**97.5%**
(Intercept)	99.821	150.912
Cancer Incidence Rate	**0.18165**	**0.20991**
Median Income	-0.00001595	0.0002902
Population	-0.000003953	0.000000914
Percent of Populace in Poverty	**0.3877**	**0.9721**
Number of Cancer-Related Clinical Trials per Capita	-0.001678	0.0009219
Median Age of Male County Residents	**-1.0616**	**-0.2552**
Median Age of Female County Residents	-0.2246	0.5571
Mean Household Size	-2.0243	1.7257
Percent of Residents who are Married	**0.61997**	**1.1924**
Percent of Married Households	**-1.1791**	**-0.63722**
Birth Rate	**-1.3815**	**-0.6312**
Percent of County Residents Aged 18-24 Highest Education: less than High School	-0.2235	-0.007226
Percent of County Residents Aged 18-24 Highest Education: High School Diploma	**0.14546**	**0.33354**
Percent of County Residents Aged 18-24 Highest Education: Bachelor's Degree	-0.23176	0.18195
Percent of County Residents Over Age 25 Highest Education: High School Diploma	**0.2005**	**0.57123**
Percent of County Residents Over Age 25 Highest Education: Bachelor's Degree	**-1.4791**	**-0.89676**
Percent of Unemployed County Residents Over Age 16	**0.27274**	**0.88274**
Percent of County Residents with Private Health Coverage	**-0.83953**	**-0.3875**
Percent of County Residents with Employee-Provided Private Health Coverage	**0.20451**	**0.56444**
Percent of County Residents with Government-Provided Health Coverage Alone	-0.22921	0.34325
Percent of County Residents who Identify as White	**-0.23463**	**-0.01151**
Percent of County Residents who Identify as Black	-0.17222	0.04282
Percent of County Residents who Identify as Asian	-0.63541	0.11816
Percent of County Residents who Identify as a Race Other than White, Black, or Asian	**-1.25198**	**-0.77293**

5 Conclusion

In this study, the data of $n = 3048$ U.S. Counties over 7 years (2010-2016) was investigated using the Multiple Linear Regression Model with the goal of predicting the Cancer Mortality Rate per 100,000 Residents in U.S. Counties based on many socioeconomic factors. Among these factors, 16 were significant at the $\alpha = 0.05$ significance level. This shows that some of the socioeconomic factors studied do have a relationship with cancer mortality rate. In order to ensure the health of our community, socioeconomic factors cannot be ignored. This model was able to predict 54% of the variability in the Cancer Mortality Rate per 100,000 Residents in U.S. Counties. This is greater than the variability that was able to be explained by any of the individual predictor variables on their own. While the variability captured is not insignificant, it would be beneficial to further investigate with more robust methods.

6 Appendix

6.1 Additional Figures

Table 8: Summary of coefficients ($\hat{\beta}$'s) of the predictor variables and their significance after removing the Median Age of County Residents predictor variable.

	Estimate ($\hat{\beta}$)	Std. Error	t value	Pr(>|t|)
(Intercept)	124.306	13.233	9.394	1.1E-20
Cancer Incidence Rate	0.192	0.007	26.658	8.6E-141
Median Income	0.000	0.000	1.139	2.5E-01
Population	-0.000	0.000	-1.460	1.4E-01
Percent of Populace in Poverty	0.564	0.153	3.683	2.3E-04
Number of Cancer-Related Clinical Trials per Capita	-0.000	0.001	-0.300	7.6E-01
Median Age of Male County Residents	-0.504	0.209	-2.410	1.6E-02
Median Age of Female County Residents	0.055	0.215	0.256	8.0E-01
Mean Household Size	0.469	0.962	0.487	6.3E-01
Percent of Residents who are Married	0.812	0.147	5.526	3.6E-08
Percent of Married Households	-0.878	0.140	-6.273	4.1E-10
Birth Rate	-0.982	0.194	-5.073	4.1E-07
Percent of County Residents Aged 18-24 Highest Education: less than High School	-0.133	0.056	-2.367	1.8E-02
Percent of County Residents Aged 18-24 Highest Education: High School Diploma	0.254	0.049	5.151	2.8E-07
Percent of County Residents Aged 18-24 Highest Education: Bachelor's Degree	-0.080	0.109	-0.737	4.6E-01
Percent of County Residents Over Age 25 Highest Education: High School Diploma	0.413	0.097	4.267	2.0E-05
Percent of County Residents Over Age 25 Highest Education: Bachelor's Degree	-1.191	0.154	-7.733	1.4E-14
Percent of Unemployed County Residents Over Age 16	0.448	0.160	2.796	5.2E-03
Percent of County Residents with Private Health Coverage	-0.543	0.131	-4.149	3.4E-05
Percent of County Residents with Employee-Provided Private Health Coverage	0.337	0.103	3.270	1.1E-03
Percent of County Residents with Government-Provided Health Coverage	0.033	0.218	0.151	8.8E-01
Percent of County Residents with Government-Provided Health Coverage Alone	0.069	0.276	0.250	8.0E-01
Percent of County Residents who Identify as White	-0.095	0.057	-1.663	9.6E-02
Percent of County Residents who Identify as Black	-0.025	0.056	-0.453	6.5E-01
Percent of County Residents who Identify as Asian	-0.005	0.189	-0.025	9.8E-01
Percent of County Residents who Identify as a Race Other than White, Black, or Asian	-0.911	0.125	-7.307	3.5E-13

Table 9: Summary of coefficients ($\hat{\beta}$'s) of the predictor variables and their significance after removing Percent of County Residents with Government-Provided Health Coverage predictor variable.

	Estimate ($\hat{\beta}$)	Std. Error	t value	Pr(>|t|)
(Intercept)	124.089	13.152	9.435	7.5E-21
Cancer Incidence Rate	0.192	0.007	26.669	6.6E-141
Median Income	0.000	0.000	1.129	2.6E-01
Population	-0.000	0.000	-1.468	1.4E-01
Percent of Populace in Poverty	0.565	0.153	3.695	2.2E-04
Number of Cancer-Related Clinical Trials per Capita	-0.000	0.001	-0.296	7.7E-01
Median Age of Male County Residents	-0.499	0.206	-2.416	1.6E-02
Median Age of Female County Residents	0.067	0.201	0.331	7.4E-01
Mean Household Size	0.466	0.962	0.484	6.3E-01
Percent of Residents who are Married	0.810	0.147	5.528	3.5E-08
Percent of County Residents Aged 18-24 Highest Education: less than High School	-0.133	0.056	-2.368	1.8E-02
Percent of County Residents Aged 18-24 Highest Education: High School Diploma	0.253	0.049	5.154	2.7E-07
Percent of County Residents Aged 18-24 Highest Education: Bachelor's Degree	-0.081	0.108	-0.750	4.5E-01
Percent of County Residents Over Age 25 Highest Education: High School Diploma	0.412	0.097	4.265	2.1E-05
Percent of County Residents Over Age 25 Highest Education: Bachelor's Degree	-1.195	0.152	-7.876	4.7E-15
Percent of Unemployed County Residents Over Age 16	0.452	0.158	2.863	4.2E-03
Percent of County Residents with Private Health Coverage	-0.534	0.116	-4.594	4.5E-06
Percent of County Residents with Employee-Provided Private Health Coverage	0.330	0.093	3.573	3.6E-04
Percent of County Residents with Government-Provided Health Coverage Alone	0.104	0.148	0.701	4.8E-01
Percent of County Residents who Identify as White	-0.095	0.057	-1.660	9.7E-02
Percent of County Residents who Identify as Black	-0.026	0.055	-0.472	6.4E-01
Percent of County Residents who Identify as Asian	-0.004	0.189	-0.022	9.8E-01
Percent of County Residents who Identify as a Race Other than White, Black, or Asian	-0.912	0.125	-7.312	3.3E-13
Percent of Married Households	-0.879	0.140	-6.286	3.7E-10
Birth Rate	-0.983	0.194	-5.075	4.1E-07

6.2 Code Chunks

```r
    setwd("C:/Users/Catherine Wu/Documents/USF/Fall 2025/STA5166/Project")
library(car)
library(reporttools)
library(xtable)
library(corrplot)
library(MASS)

## Data

# Read in the data set
cancer <- read.csv("cancer_reg.csv", header = TRUE)

# Histogram of cancer mortality rates
hist(cancer$TARGET_deathRate,
     xlab = "Cancer Mortality Rates per 100,000 People",
     main = "",
     col = "gray")

# Make a table of descriptive statistics
descstats <- summary(cancer)
contcancer <- cancer[, !(names(cancer) %in% c("binnedInc", "Geography", "avgAnnCount", "avgDeathsPerYear", "PctSomeCol18_24", "PctEmployed16_Over", "PctPrivateCoverageAlone"))]
tableContinuous(
  vars = contcancer,
  stats = c("mean", "median", "min", "max"),
  cap = "Descriptive Statistics of Socioeconomic Factors Affecting Cancer Mortality",
  lab = "tab:descriptivestats"
)

## Exploratory Data Analysis Grouped by Related Predictor Variables

# Correlation Matrix Heatmap
corrplot(cor(contcancer),
         method = "color",
         col = colorRampPalette(c("blue", "white", "red"))(100),
         tl.col = "black",
         tl.srt = 60,
         cl.pos = "r",
         cl.cex = 1,
         cl.ratio = 0.2)

# Removed: Some College, Employment Rate, and Private Coverage Alone are missing values

# Initial Model
g1 = lm(TARGET_deathRate ~ incidenceRate + medIncome + popEst2015 + povertyPercent
        + studyPerCap + MedianAge + MedianAgeMale + MedianAgeFemale + AvgHouseholdSize
        + PercentMarried + PctNoHS18_24 + PctHS18_24 + PctBachDeg18_24 + PctHS25_Over
        + PctBachDeg25_Over + PctUnemployed16_Over + PctPrivateCoverage
        + PctEmpPrivCoverage + PctPublicCoverage + PctPublicCoverageAlone + PctWhite
        + PctBlack + PctAsian + PctOtherRace + PctMarriedHouseholds + BirthRate,
        data = cancer)
```

```r
summary(g1)

# ANOVA Test of Initial Model with Reduced Model
g0 = lm(TARGET_deathRate~1, data = cancer)
anova(g1, g0)

# Export table of initial model to Latex
digits_matrix <- matrix(3, nrow = nrow(summary(g1)$coefficients), ncol = ncol(summary(g1)$coefficients)+1)
digits_matrix[, 5] <- -1
g1tab <- xtable(g1, digits = digits_matrix)
print(g1tab)

## Model Diagnostics of Initial Model

# Generate Residual Plot
plot(summary(g1)$residuals,
     main = "",
     xlab = "",
     ylab = "Residuals",
     pch = 20,
     ylim = c(-150,150))
abline(h = 0, lty = 2)

# Generate Studentized Residual Plot
plot(studres(g1),
     main = "",
     xlab = "",
     ylab = "Studentized Residuals",
     pch = 20,
     ylim = c(-7, 7))
abline(h = 0, lty = 2)

# Plot the Residuals vs. the Fitted Values
plot(fitted.values(g1),
     residuals(g1),
     main = "",
     xlab = expression(hat(y)),
     ylab = "Residuals",
     pch = 20)

# Plot the Normal Plot
qqnorm(residuals(g1),
       pch = 20,
       main = "",
       xlab = "Normal Scores",
       ylab = "Ordered Residuals")

# Remove MedianAge due unusual values and insignificant contribution to the model
gt = lm(TARGET_deathRate ~ incidenceRate + medIncome + popEst2015 + povertyPercent
        + studyPerCap + MedianAgeMale + MedianAgeFemale + AvgHouseholdSize
        + PercentMarried + PctNoHS18_24 + PctHS18_24 + PctBachDeg18_24 + PctHS25_Over
        + PctBachDeg25_Over + PctUnemployed16_Over + PctPrivateCoverage
        + PctEmpPrivCoverage + PctPublicCoverage + PctPublicCoverageAlone + PctWhite
        + PctBlack + PctAsian + PctOtherRace + PctMarriedHouseholds + BirthRate,
        data = cancer)
summary(gt)
```

```

# Export table of model with MedianAge removed to Latex
digits_matrix <- matrix(3, nrow = nrow(summary(gt)$coefficients), ncol = ncol(
    summary(gt)$coefficients)+1)
digits_matrix[, 5] <- -1
gttab <- xtable(gt, digits = digits_matrix)
print(gttab)

# ANOVA Test to Check Significance of Removing MedianAge
anova(gt, g1)

# Check for multiocollinearity and adjust the model
vif(gt)

# Remove PctPublicCoverage due to high VIF and insignificant contribution to the
    model
gt2 = lm(TARGET_deathRate ~ incidenceRate + medIncome + popEst2015 +
    povertyPercent
          + studyPerCap + MedianAgeMale + MedianAgeFemale + AvgHouseholdSize
          + PercentMarried + PctNoHS18_24 + PctHS18_24 + PctBachDeg18_24 + PctHS25_
              Over
          + PctBachDeg25_Over + PctUnemployed16_Over + PctPrivateCoverage
          + PctEmpPrivCoverage + PctPublicCoverageAlone + PctWhite
          + PctBlack + PctAsian + PctOtherRace + PctMarriedHouseholds + BirthRate,
          data = cancer)
summary(gt2)

# Export table of model with PctPublicCoverage removed to Latex
digits_matrix <- matrix(3, nrow = nrow(summary(gt2)$coefficients), ncol = ncol(
    summary(gt2)$coefficients)+1)
digits_matrix[, 5] <- -1
gt2tab <- xtable(gt2, digits = digits_matrix)
print(gt2tab)

# ANOVA Test to Check Significance of Removing PctPublicCoverage
anova(gt2, gt)

vif(gt2)

## Detect and Remove Outliers from the Multicollinearity-adjusted Model

# Find the influential observations (leverage values)
gt2inf = influence(gt2)
sum(gt2inf$hat)
infindt2 = seq(1, length(gt2inf$hat))[gt2inf$hat > (2*(length(coef(gt2)))/length(
    gt2inf$hat))]

# Find the estimate of the standard deviation of the errors
summary(gt2)$sig

# Find the outliers using studentized residuals
studres = residuals(gt2)/(summary(gt2)$sig*sqrt(1-gt2inf$hat))
sroutt2 = seq(1, length(studres))[abs(studres) > 3]

# Remove data points that are influential observations and outliers
b = intersect(infindt2, sroutt2)
cancertest_clean <- cancer[-b, ]

# Model with Outliers Removed
```

```r
gtc = lm(TARGET_deathRate ~ incidenceRate + medIncome + popEst2015 + povertyPercent
         + studyPerCap + MedianAgeMale + MedianAgeFemale + AvgHouseholdSize
         + PercentMarried + PctNoHS18_24 + PctHS18_24 + PctBachDeg18_24 + PctHS25_Over
         + PctBachDeg25_Over + PctUnemployed16_Over + PctPrivateCoverage
         + PctEmpPrivCoverage + PctPublicCoverageAlone + PctWhite
         + PctBlack + PctAsian + PctOtherRace + PctMarriedHouseholds + BirthRate,
         data = cancertest_clean)
summary(gtc)

# Export table of model with outliers removed to Latex
digits_matrix <- matrix(3, nrow = nrow(summary(gtc)$coefficients), ncol = ncol(summary(gtc)$coefficients)+1)
digits_matrix[, 5] <- -1
gtctab <- xtable(gtc, digits = digits_matrix)
print(gtctab)

# ANOVA Test of Model with Outliers Removed and Reduced Model
gt0 = lm(TARGET_deathRate~1, data = cancertest_clean)
anova(gtc, gt0)

## Model Diagnostics of Final Model

# Generate Residual Plot
plot(summary(gtc)$residuals,
     main = "",
     xlab = "",
     ylab = "Residuals",
     pch = 20,
     ylim = c(-150,150))
abline(h = 0, lty = 2)

# Generate Studentized Residual Plot
plot(studres(gtc),
     main = "",
     xlab = "",
     ylab = "Studentized Residuals",
     pch = 20,
     ylim = c(-7, 7))
abline(h = 0, lty = 2)

# Plot the Residuals vs. the Fitted Values
plot(fitted.values(gtc),
     residuals(gtc),
     main = "",
     xlab = expression(hat(y)),
     ylab = "Residuals",
     pch = 20)

# Plot the Normal Plot
qqnorm(residuals(gtc),
       pch = 20,
       main = "",
       xlab = "Normal Scores",
       ylab = "Ordered Residuals")

## Building Confidence Intervals for the True Values of Regression Coefficients
confint(gtc)
```

6.3 References

References

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